

Transcorrelated Jastrow Fitting: Atomization Energies

1 Theory

The selected virtual space is constructed by projecting the large-basis virtual MOs onto the small-basis AO space via

$$M = S_{11}^{-1/2} S_{12} C^{virt} \quad [n_{small} \times n_{large,virt}] \quad (1)$$

where S_{11} and S_{12} are the small-basis and cross-overlap matrices, and C^{virt} collects the large-basis virtual MO coefficients. The elements $M_{\mu a} = \langle \tilde{\varphi}_\mu | \psi_a \rangle$ are overlaps between Löwdin-orthonormalized small AOs and S_{22} -orthonormal large-basis virtuals, so the thin SVD

$$M = U \Sigma V^\top \quad U [n_{small} \times n_{small}], \Sigma [n_{small} \times n_{small}], V [n_{large,virt} \times n_{small}] \quad (2)$$

yields singular values $\sigma_k \leq 1$ that are exact cosines of the principal angles between the two subspaces. Dropping the n_{occ} columns of V with near-zero σ_k and projecting onto the AO basis gives the $n_{sel} = n_{small} - n_{occ}$ selected virtuals

$$C^{sel} = C^{virt} V_{kept} \quad [n_{large,AO} \times n_{sel}] \quad (3)$$

which are S_{22} -orthonormal by construction. The SVD rotation mixes canonical virtuals, rendering the projected Fock matrix

$$F^{sel} = V_{kept}^\top \text{diag}(\varepsilon_a) V_{kept} \quad [n_{sel} \times n_{sel}] \quad (4)$$

non-diagonal. Diagonalizing $F^{sel} = W \tilde{\Lambda} W^\top$ yields the semicanonical orbitals

$$C^{final} = C^{virt} V_{kept} W \quad [n_{large,AO} \times n_{sel}] \quad (5)$$

satisfying $\langle \hat{a}_m | \hat{F} | \hat{a}_n \rangle = \tilde{\varepsilon}_m \delta_{mn}$, with orbital energies $\tilde{\varepsilon}_m$ entering the CCSD(T) denominators $D_{ij}^{ab} = \varepsilon_i + \varepsilon_j - \tilde{\varepsilon}_a - \tilde{\varepsilon}_b$.

2 Atomization energies

We compute atomization energies with several xTC method families and benchmark them against SHCI+PBE+CV reference values. All calculations employ effective core potentials (ECPs) for the third-row elements. Per-basis, per-molecule method discrepancies and the full basis set convergence of error metrics are hosted as interactive plotly figures [1].

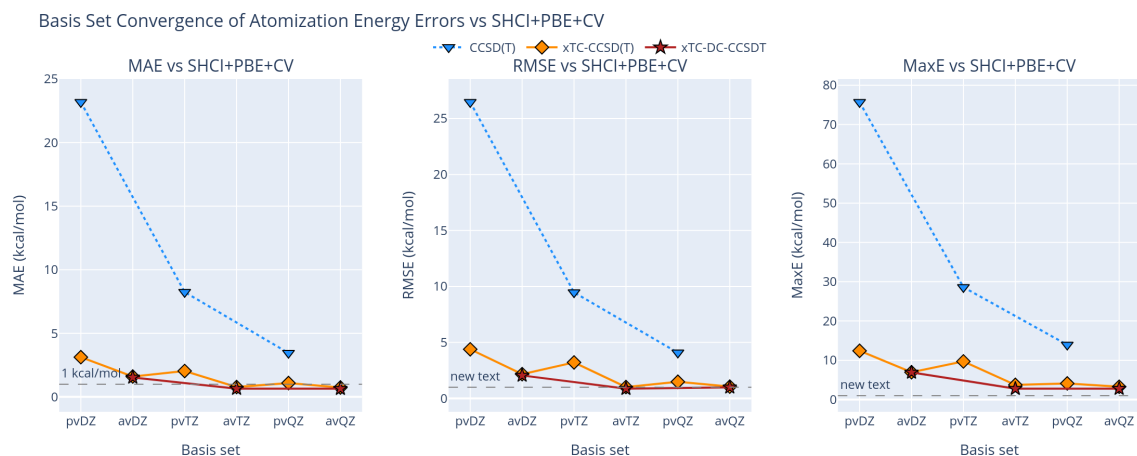


Figure 1: Basis set convergence (MAE, RMSE, MaxE in kcal/mol) of atomization energy errors vs SHCI+PBE+CV for each method family, on a unified x-axis interleaving cc-pVxZ and aug-cc-pVxZ. The interactive version and per-basis per-molecule breakdowns are available online [1].

References

- [1] Kristoffer Simula. Interactive plotly figures: xtc atomization energies vs shci+pbe+cv. Online collection, GitHub Pages, 2026. <https://simulak1.github.io/tc-g2-data/>.

Table 1: MAE, RMSE, and MaxE of atomization energies vs SHCI+PBE+CV reference (kcal/mol), for each method and basis set.

Method	Basis	MAE	RMSE	MaxE
(kcal/mol)				
<i>SHCI</i>				
	CBS	0.18	0.26	0.75
<i>xTC-CCSD(T)</i>				
	pvDZ	3.13	4.38	12.40
	avDZ	1.60	2.16	6.95
	pvTZ	2.04	3.21	9.70
	avTZ	0.78	1.01	3.73
	pvQZ	1.09	1.48	4.10
	avQZ	0.74	1.05	3.26
<i>xTC-DC-CCSDT</i>				
	avDZ	1.53	2.05	6.90
	avTZ	0.65	0.87	2.77
	avQZ	0.65	0.99	2.78
<i>CCSD(T)</i>				
	pvDZ	23.18	26.47	75.71
	pvTZ	8.25	9.46	28.60
	pvQZ	3.45	4.08	13.94
	pv5Z	1.67	1.96	5.52